Students will be assessed on their ability to:		Suggested experiments
91	explore and use the terms magnetic flux density B, flux Φ and flux linkage $N\Phi$	
92	investigate, recognise and use the expression $F = BIl \sin\theta$ and apply Fleming's left hand rule to currents	Electronic balance to measure effect of I and l on force
94	investigate and explain qualitatively the factors affecting the emf induced in a coil when there is relative motion between the coil and a permanent magnet and when there is a change of current in a primary coil linked with it	Use a data logger to plot V against t as a magnet falls through a coil of wire
95	investigate, recognise and use the expression $\varepsilon = -\mathrm{d}(N\Phi)/\mathrm{d}t$ and explain how it is a consequence of Faraday's and Lenz's laws	

7.4 The Medium is the Message (MDM)

Students will learn about the physics involved in some modern communication and display techniques:

- fibre optics and exponential attenuation
- CCD imaging
- cathode ray tube
- liquid crystal and LED displays.

Exponential functions are used to model attenuation losses.

There are opportunities for students to develop information and communication technology skills using computer simulations.

In studying various types of display technology, students will consider their relative power demands and discuss the choices made by organisations and by individuals.

Students will be assessed on their ability to:		Suggested experiments
83	explain what is meant by an electric field and recognise and use the expression electric field strength $E = F/Q$	
84	draw and interpret diagrams using lines of force to describe radial and uniform electric fields qualitatively	Demonstration of electric lines of force between electrodes
86	investigate and recall that applying a potential difference to two parallel plates produces a uniform electric field in the central region between them, and recognise and use the expression $E = V/d$	
87	investigate and use the expression $C = Q/V$	Use a Coulometer to measure charge stored
88	recognise and use the expression $W = ½QV$ for the energy stored by a capacitor, derive the expression from the area under a graph of potential difference against charge stored, and derive and use related expressions, for example, $W = ½CV^2$	Investigate energy stored by discharging through series/parallel combination of light bulbs
91	explore and use the terms magnetic flux density B, flux Φ and flux linkage $N\Phi$	